Multiple Choice Questions (MCQ)

➢ Temperature and Thermal Expansion

1. **The temperature at which the volume of a gas becomes zero is called**
   a) The absolute scale of temperature
   b) The absolute zero temperature
   c) The absolute temperature
   d) None of these

2. **When two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other. This statement is called**
   a) Zeroth law of thermodynamics
   b) First law of thermodynamics
   c) Second law of thermodynamics
   d) Kelvin Planck's law

3. **The scales of temperature are based on two fixed points which are**
   a) The temperatures of water at 0° 100°C
   b) The temperature of melting ice and boiling water at atmospheric pressure
   c) The temperatures of ice cold and boiling water
   d) The temperatures of frozen and boiling mercury

4. **The absolute zero temperature is taken as**
   a) -273°C
   b) 273°C
   c) 237°F
   d) -237°F

5. **The temperature of a body at 100°C is increased by ΔT as measured on the Celsius scale. How is this temperature change expressed on the Kelvin scale?**
   a) ΔT
   b) ΔT + 273
   c) ΔT – 273
   d) ΔT + 373

6. **A temperature difference of 5 K is equal to:**
   a) A difference of 9 on the Celsius scale.
   b) A difference of 9 on the Fahrenheit scale.
   c) A difference of 5 on the Fahrenheit scale.
   d) None of these

7. **A temperature difference of 9.0 Celsius degrees is equal to a Fahrenheit temperature difference of**
   a) 5.0 Fahrenheit degrees.
   b) 9.0 Fahrenheit degrees.
   c) 16 Fahrenheit degrees.
   d) 37 Fahrenheit degrees.

8. **A material of length (L) at temperature (T_1) K is subjected to a temperature rise of (T) K. The coefficient of linear expansion of the material is (α) K^{-1}. The material expands by:**
   a) $L(1 + \alpha T)$
   b) $L\,\alpha(T - T_1)$
   c) $L\,[1 + \alpha(T - T_1)]$
   d) $L\,\alpha T$

9. **When the temperature of a rod of copper is increased, its length:**
   a) stays the same
   b) increases
   c) decreases
   d) None of these

10. **The amount by which unit length of a material increases when the temperature is raised one degree is called the coefficient of:**
   a) cubic expansion
   b) superficial expansion
   c) linear expansion
   d) volume expansion

11. **A building made with a steel structure is 650 m high on a winter day when the temperature is 0° F. How much taller (in cm) is the building when it is 100° F? (The linear expansion coefficient of steel is 11 x 10^{-6} °C^{-1}.)**
   a) 71
   b) 36
   c) 40
   d) 46

12. **A gold ring has an inner diameter of 2.168 cm at a temperature of 15.0°C. Determine its diameter at 100.0°C. (α_{GOLD} = 1.42 x 10^{-5} °C^{-1})**
   a) 2.171cm
   b) 2.171 m
   c) 0.0217 cm
   d) None of the above

13. **When a body is heating, one can note that the relation between the Volume expansion coefficient (β) and the linear expansion coefficient (α) given by**
   a) $\beta = 2\alpha$
   b) $\alpha = 3\,\beta$
   c) $\alpha = 2\,\beta$
   d) $\beta = 3\,\alpha$

14. **When a body is heating, one can note that the relation between the Area expansion coefficient (γ) and the Volume expansion coefficient (B) given by**
   a) $\gamma = 1.5\,\beta$
   b) $\beta = 2\,\gamma$
   c) $\gamma = 0.67\,\beta$
   d) $\gamma = 2\,\beta$

➢ First Law of Thermodynamics

15. The SI unit of specific heat capacity is

a) Jkg^{-1}
b) K^{-1}
c) $Jkg^{-1}\ {}^{o}C^{-1}$
d) Pa

15. The amount of heat required to raise the temperature of 1kg by 1°C is called

a) specific energy
b) specific heat capacity
c) specific stress
d) specific density

16. ……. is the amount of heat energy required to change a substance from solid to liquid at its melting point, without changing its

a) Latent heat of Fusion
b) Latent Heat of Vaporization
c) Heat Capacity
d) None of these

17. A 5-kg piece of lead (specific heat capacity 0.03 cal/g °C) having a temperature of 80°C is added to 500 g of water having a temperature of 20°C. What is the final equilibrium temperature (in °C) of the system?

a) 79
b) 26
c) 54
d) 34

18. Unit of the rate of heat transfer is

a) Joule
b) Newton
c) Pascal
d) Watt

19. The main mode of heat transfer through fluids is:

a) Conduction
b) Convection
c) Radiation
d) None of the above

20. What is unit of the heat transfer coefficient of fluid?

a) $W/m^2\ k$
b) W/m
c) W/m k
d) W/m^3

21. Amount of heat flow through a substance by conduction is.....

a) Depend upon on the material of substance
b) Directly proportional to temperature difference of substance
c) Directly proportional to surface area of substance
d) All of the above

22. The thermal resistance of a single homogeneous solid is increased if.

a) The thermal conductivity is increased.
b) The cross-sectional area is increased.
c) The thickness of the slab is increased.
d) All of the above.

23. According to the non-flow equation, the heat flow during the process of constant Volume will equal the change:

a) In internal energy.
b) In enthalpy.
c) In entropy.
d) In temperature.

24. The work done by a gas during expansion or compression is equal to:

a) The change in pressure.
b) The change in volume.
c) The area under the ***P-V*** diagram.
d) The change in temperature

25. Due to the first law of thermodynamic, when systems undergoes thermodynamics cycle

a) Δh_t = zero
b) $W_t = Q_t$
c) Q_t = zero
d) a and b are correct

26. According to the non-flow equation, the work done will not equal the heat flow due to the changing

a) In internal energy.
b) In enthalpy.
c) In entropy.
d) In temperature.

27. When the gas is heated at constant volume

a) Its temperature will increase
b) Its pressure will increase
c) Both temperature and pressure will increase
d) Neither temperature nor pressure will increase

28. The unit of the specific volume is:

a) $\frac{m^3}{Kg\ c^0}$

b) $\frac{m^3}{c^0}$

c) $\frac{m^3}{Kg}$

d) $\frac{Joule}{kg}$

29. When the gas is cooled at constant pressure,

a) Its temperature increases but volume decreases
b) Its volume increases but temperature decreases
c) Both temperature and volume increases
d) Both temperature and volume decreases

30. In the isochoric process

a) The temperature remains constant.
b) The pressure remains constant.
c) The volume remains constant.
d) The internal energy is constant.

31. A gas is not able to do work under the ……..

a) Isobaric process.
b) Isothermal process.
c) Isochoric process.
d) Adiabatic process

32. The ratio of specific heat at constant pressure (cp) and specific heat at constant volume (cv) is always _____ one.

a) equal to
b) less than
c) greater than
d) negative

33. The general gas equation for perfect gas is (where p = Pressure, V = Specific Volume, m = mass, T = Absolute temperature, and R = Gas constant)

a) PV = mRT
b) PV = RT^m
c) PV=RT
d) PV = $(RT)^m$

34. The value of one bar (in S. I. units) is equal to

a) 1×10^2 N/m^2
b) 1×10^3 N/m^2
c) 1×10^4 N/m^2
d) 1×10^5 N/m^2

35. The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume, is called

a) Specific heat capacity at constant volume
b) Specific heat capacity at constant pressure
c) Specific heat capacity at constant temperature
d) kilo Joule

36. The specific gas constant (R) is equal to the…… between the specific heat capacity at constant pressure and the specific heat capacity at constant volume

a) Sum
b) Difference
c) Product
d) Ratio

37. If specific heat ratio for a gas is (ɣ), what is the change in internal energy of a mass of gas at constant pressure P, when volume change from V to 2V is,

a) $\frac{PV}{\gamma-1}$
b) $\frac{R}{\gamma-1}$
c) PV
d) $\frac{\gamma PV}{\gamma-1}$

38. Specific heat capacity at constant volume is,

a) The ratio of the change in specific enthalpy to the corresponding change in temperature.
b) The ratio of the change in specific internal energy to the corresponding change in temperature.
c) The ratio of the change in pressure to the corresponding change in volume.
d) The ratio of the change in volume to the corresponding change in pressure.

39. For perfect gas, which of the following is correct in isothermal process

a) P is constant
b) V is constant
c) PV is constant
d) P/T is constant

40. The heat flow to or from the system is zero if the process is:

a) Isobaric.
b) Isothermal.
c) Constant volume.
d) Adiabatic.

41. For perfect gas, the heat flow to or from the system equals the work done if the process is:

a) Polytropic.
b) Isothermal.
c) Isochoric.
d) Adiabatic

➢ Heat Engine and the second law of thermodynamics

42. Which of the following is the Kelvin–Planck statement of the second law of thermodynamics for a heat engine?

a) A heat engine can't turn all heat into work in a cycle.
b) Heat can't flow from cold to hot reservoir without work.
c) A heat engine can run with zero heat loss.
d) None of these.

43. Which of the following refers to the term C.O.P. of refrigeration?

a) Cooling for Performance.
b) Coefficient of Performance.
c) Capacity of Performance.
d) Co-efficient of Plant

44. What is the definition of the coefficient of performance (COP) for a refrigerator?

a) The ratio of work input to heat removed.
b) The ratio of heat removed to the work input.
c) The ratio of total heat to work output.
d) The ratio of work input to total energy input.

45. The thermal efficiency of the heat engine is always

a) Higher than 1
b) Lower than 1
c) Equal 1
d) Equal zero

46. The second process in the Carnot cycle is

a) Isentropic expansion
b) Isentropic compression
c) Isothermal expansion
d) Isothermal compression

47. The Condition of the Carnot cycle say that the ratio of the energy transferee by heat between two reservoirs is equal to the ratio of the reservoir

a) Temperatures
b) Pressures
c) Volumes
d) None of this.

48. Two Carnot engines, A and B, operate in succession. Engine A takes in heat from a 1600 K reservoir and releases heat to an intermediate reservoir at temperature T. Engine B absorbs this rejected heat and further releases it to a low temperature reservoir at 450 K. Find the temperature T (in kelvin) if the ratio of the efficiencies of engine A to engine B is 2:1?

a) 800 K.
b) 643 K.
c) 1000 K.
d) 750 K.

49. A cyclic heat engine does 24 KJ of work per cycle. If the efficiency of the engine is 0.8, then what will be the heat in KJ rejected per cycle?

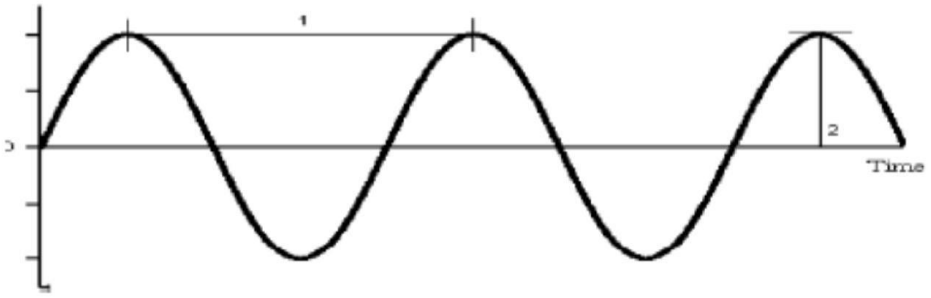
a) 30
b) 6
c) 19.2
d) 39.2

➢ Wave Motion

50. Which of the following quantities is transferred during wave propagation?

a) Speed
b) Mass
c) Matter
d) Energy

51. Refer to the sound wave below, Numbers 1 and 2, respectively, correspond to----------



a) frequency and wavelength.
b) wavelength and frequency.
c) wavelength and amplitude.
d) amplitude and wavelength.

52. The distance from one wave crest to the next wave crest is called

a) Wavelength
b) Amplitude
c) Frequency
d) Periodic time

53. The number of vibrations per unit time is called the

a) Wavelength.
b) Period.
c) Frequency
d) Node

54. The particles of the medium vibrate in longitudinal waves …

a) perpendicular to the direction of the wave motion
b) Along the direction of wave motion
c) Opposite to the direction of the wave motion
d) Do not vibrate at all

55. The waves in which material medium is required for their propagation are known as

a) Mechanical waves
b) Electromagnetic waves
c) All of these
d) None of these

56. The amplitude of a wave is ----------

a) The distance the wave moves in one second.
b) The distance the wave moves in one-time period of the wave.
c) The maximum distance moved by the medium particles on either side of the mean position.
d) The distance equal to one wavelength.

57. Sound waves are ----------

a) Longitudinal
b) Transverse
c) Partly longitudinal and partly transverse
d) Sometimes longitudinal and sometimes transverse

58. Sound waves do not travel through ----------

a) Solids
b) Liquids
c) Gases
d) Vacuum

59. The speed of sound in medium depends upon

a) Amplitude
b) Frequency
c) Wavelength
d) properties of the medium

60. Frequencies of sound that are too high for the human ear to hear are called

a) supersonic
b) infrasonic
c) ultrasonic
d) subsonic

61. About what frequency range can the average human hear?
a) 2Hz to 200Hz
b) 200Hz to 2000Hz
c) 20Hz to 20,000Hz
d) 200Hz to 200,000Hz

62. Above what frequency is ultrasound?
a) 20Hz
b) 2000Hz
c) 20,000Hz
d) 200,000Hz

63. When a sound wave passes from air into water, which of the following does not change?
a) speed
b) frequency
c) wavelength
d) none of the above

64. Which state of matter do sounds travel fastest in?
a) Solid
b) Liquid
c) Gas
d) none of the above

65. What is the unit of measure for the loudness of a sound?
a) Amps
b) Decibels
c) Hertz
d) Meters

66. Which of the following represents the loudest sound?
a) 262 Hz and 120 dB
b) 330 Hz and 100 dB
c) 415 Hz and 50 dB
d) 440 Hz and 20 dB

67. What is the decibel level of a sound that has an intensity of 1.0 W/m², if the threshold of hearing is $1x10^{-12}$ W/m²?
a) 0 dB
b) 70 dB
c) 120 dB
d) 140 dB

68. The intensity of a sound at any distance from the source is directly proportional to the sound's
a) wavelength
b) pitch
c) power
d) frequency

69. At a distance of 3 m, the intensity of a sound will be ____ the intensity it was at a distance of 1m.

a) one-ninth
b) 9 times
c) one-third
d) 3 times

70. The Doppler Effect occurs with

a) only sound waves.
b) only transverse waves.
c) only water waves.
d) all waves.

71. When a police car comes towards you, the sound of the siren will become_____

a) high pitched
b) lower pitched
c) polarized.
d) there is not enough information to tell.

72. Suppose that a train whistle at rest has a frequency of 3000 Hz. If you are standing still and observe the frequency to be 3010 Hz (higher), you can conclude that

a) the train is moving away from you.
b) the train is moving toward you
c) the sound from the whistle has echoed
d) not enough information is given

73. A train moves away from you at a speed of 50.0m/s. If you are standing still and hear the whistle at a frequency of 305Hz, what is the actual frequency of the produced whistle? (Assume speed of sound to be 331m/s.)

a) 259Hz
b) 205Hz
c) 405Hz
d) 351Hz

74. A girl moves away from a source of sound at a constant speed. Compared to the frequency of the sound wave produced by the source, the frequency of the sound wave heard by the girl is

a) lower.
b) higher.
c) the same.
d) None of these.